

Rewriting Circles and Parabolas in Standard Form

Completing the square on parabolas and circles, reading features off standard form, and building an equation from given features

- Standard forms: a parabola is $y = a(x - h)^2 + k$; a circle is $(x - h)^2 + (y - k)^2 = r^2$.
 - To complete the square on $x^2 + px$, add $\left(\frac{p}{2}\right)^2$ — and keep the equation balanced by adding it to the other side too (or subtracting it back on the same side).
 - Factor out any leading coefficient before completing the square, and put it back afterwards.
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1. Rewrite $y = x^2 + 6x + 5$ in the form $y = (x - h)^2 + k$.

Answer: _____

2. Rewrite $x^2 + y^2 - 4x - 12 = 0$ in the form $(x - h)^2 + (y - k)^2 = r^2$.

Answer: _____

3. Rewrite $y = x^2 - 10x + 18$ in the form $y = (x - h)^2 + k$.

Answer: _____

4. Rewrite $x^2 + y^2 + 8x - 6y + 9 = 0$ in the form $(x - h)^2 + (y - k)^2 = r^2$.

Answer: _____

5. Consider the circle $x^2 + y^2 - 6x + 4y - 12 = 0$.

(a) Rewrite it in the form $(x - h)^2 + (y - k)^2 = r^2$.

Answer: _____

(b) Write down the radius.

Answer: _____

6. Rewrite $y = 2x^2 - 12x + 13$ in the form $y = a(x - h)^2 + k$. Factor the leading coefficient out first.

Answer: _____

7. Consider the parabola $y = x^2 + 5x + 3$. The constant to complete the square is not a whole number here; leave every value as an exact fraction.

(a) Rewrite the equation in the form $y = (x - h)^2 + k$.

Answer: _____

(b) Write the x -coordinate of the vertex.

Answer: _____

8. Rewrite $2x^2 + 2y^2 - 12x + 20y + 18 = 0$ in the form $(x - h)^2 + (y - k)^2 = r^2$. Divide through by the common factor before completing either square.

Answer: _____

9. A circle has centre $(-4, 7)$ and radius 6. Write its equation in standard form, then multiply out and simplify to the general form $x^2 + y^2 + Dx + Ey + F = 0$.

Answer: _____

10. A parabola has vertex $(2, -9)$ and passes through the point $(5, 0)$. Its equation has the form $y = a(x - 2)^2 - 9$.

(a) Substitute the point $(5, 0)$ and solve for a .

Answer: _____

(b) Multiply out your equation and simplify it to the form $y = x^2 + bx + c$.

Answer: _____

11. Consider the parabola $y = 3x^2 + 12x + 7$.

(a) Rewrite it in the form $y = a(x - h)^2 + k$.

Answer: _____

(b) Write the y -coordinate of the vertex.

Answer: _____

12. Consider $x^2 + y^2 - 2x + 8y + 8 = 0$.

(a) Rewrite it in the form $(x - h)^2 + (y - k)^2 = r^2$.

Answer: _____

(b) Write down the radius.

Answer: _____

(c) Explain how the original equation already rules out a parabola, before any square is completed.

Answer: _____